

Problem statement

You are given a tiny attention-based sequence classifier in PyTorch. The script runs, but the model underperforms and some metrics are misleading.

Your tasks:

1. Fix the code so training is correct and stable.
2. Reach **validation accuracy ≥ 0.83** within **8 epochs** on CPU.
3. Add two quick dataset checks:
 - training positive-class rate
 - mean valid sequence length
4. After training, print the **top 5 validation examples with highest loss** and explain whether some may be noisy labels.
5. Answer the short conceptual questions after the coding section.

What this tests:

- debugging ML code
- vectors/matrices and attention masking
- PyTorch autograd and optimization
- data quality / noisy labels
- evaluation strategy